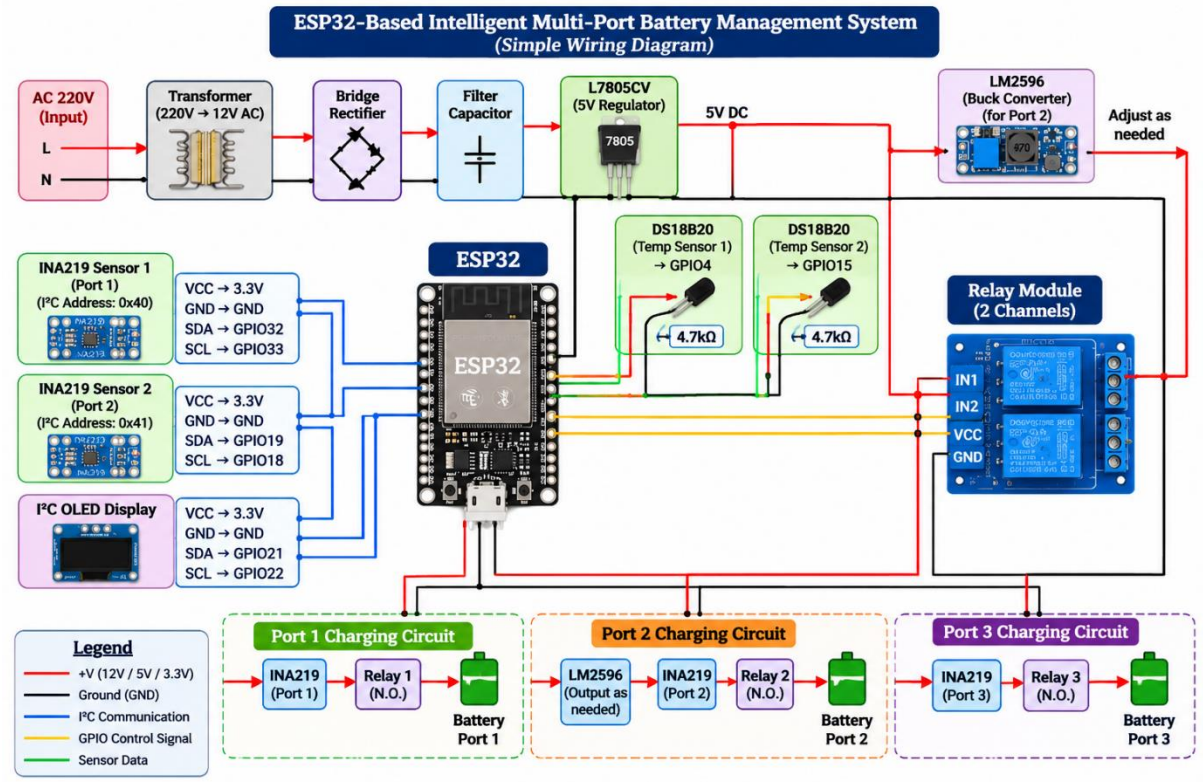
**Complete Hardware Connection:**




**1. Main AC-to-DC power supply**

| Component | Pin / Terminal | Connection |
|---|---|---|
| **AC mains** | Live (L) | Transformer primary L |
| **AC mains** | Neutral (N) | Transformer primary N |
| **Transformer** | Secondary AC1 | Bridge rectifier (~) |
| **Transformer** | Secondary AC2 | Bridge rectifier (~) |
| **Bridge rectifier** | (+) | Filter capacitor (+) |
| **Bridge rectifier** | (−) | Filter capacitor (−) |
| **Filter capacitor** | (+) | DC positive supply |
| **Filter capacitor** | (−) | Common DC GND |
| **L7805CV** | IN | Filtered DC positive |
| **L7805CV** | GND | Common DC GND |
| **L7805CV** | OUT | +5V relay supply |

**2. ESP32 connections**

| ESP32 Pin | Connected Component | Component Pin |
|---|---|---|
| **3.3V** | INA219 #1 | VCC |

| | | |
|---|---|---|
| **3.3V** | INA219 #2 | VCC |
| **3.3V** | DS18B20 #1 | VDD |
| **3.3V** | DS18B20 #2 | VDD |
| **GPIO 32** | INA219 #1 | SDA |
| **GPIO 33** | INA219 #1 | SCL |
| **GPIO 19** | INA219 #2 | SDA |
| **GPIO 18** | INA219 #2 | SCL |
| **GPIO 4** | DS18B20 #1 | DATA |
| **GPIO 15** | DS18B20 #2 | DATA |
| **GPIO 12** | Relay module | IN1 |
| **GPIO 14** | Relay module | IN2 |
| **GPIO 27** | Relay module | IN3 |
| **GPIO 26** | Relay module | IN4 |
| **GND** | All low-voltage modules | GND |

**3. DS18B20 temperature sensors**

| DS18B20 Pin | Connection |
|---|---|
| **VDD (Red)** | ESP32 3.3V |
| **GND (Black)** | Common GND |
| **DATA (Yellow) Sensor 1** | GPIO 4 |
| **DATA (Yellow) Sensor 2** | GPIO 15 |
| **4.7 kΩ resistor #1** | GPIO 4 DATA to 3.3V |
| **4.7 kΩ resistor #2** | GPIO 15 DATA to 3.3V |

**4. Relay module connections**

| Relay Module Pin | Connection |
|---|---|
| **VCC** | Regulated +5V |
| **GND** | Common GND |
| **IN1** | ESP32 GPIO 12 |
| **IN2** | ESP32 GPIO 14 |
| **IN3** | ESP32 GPIO 27 |
| **IN4** | ESP32 GPIO 26 |

**5. INA219 current sensor connections**

| INA219 #1 | Connection |
|---|---|
| **VCC** | ESP32 3.3V |
| **GND** | Common GND |
| **SDA** | GPIO 32 |
| **SCL** | GPIO 33 |
| **VIN+** | Port 1 supply positive |
| **VIN−** | Port 1 battery positive |

| INA219 #2 | Connection |
|---|---|
| **VCC** | ESP32 3.3V |
| **GND** | Common GND |
| **SDA** | GPIO 19 |
| **SCL** | GPIO 18 |
| **VIN+** | LM2596 OUT+ |
| **VIN−** | Port 2 battery positive |

**6. LM2596 DC-DC buck converter (Port 2)**

| LM2596 Terminal | Connection |
|---|---|
| **IN+** | Relay 2 NO2 |
| **IN−** | Common DC GND |
| **OUT+** | INA219 #2 VIN+ |
| **OUT−** | Common DC GND |
| **Adjust potentiometer** | Set required charging voltage before connecting battery |